

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        // int i = 1; // init
        // int m = 5;
        // while(i <= m){ // condition
        //     // self work
        //     System.out.println(i);
        //     i++; //variable increment
        // }
```

```
        // int i = 5; // init
        // int m = 1;
        // while(i >= m){ // condition
        //     // self work
        //     System.out.println(i);
        //     i--; //variable increment
        // }
```

```
        // // Way-1
        // int m = 5;
        // for(int i=1; i <= m; i++){
        //     // self work
        //     System.out.println(i);
        // }
```

```
        // // Way-2
        // int m = 5;
        // int i=1;
        // for( i <= m; i++){
        //     // self work
        //     System.out.println(i);
        // }
```

```
        // // Way-3
        // int m = 5;
        // int i=1;
        // for( i <= m; ){
        //     // self work
        //     System.out.println(i);
        //     i++;
        // }
```

```
        int j = 2;
        int i = 2;
        int m = 5;
        for( ; i <= m; j++){
            System.out.println(j);
        }
```

```
    }
```

```
}
```

Prime numbers

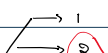
16 → ? → 1, 2, 4, 8, 16

A no. which is divisible by 1 or itself.

$n = 10$

$n = 10$, it is divisible by 5.

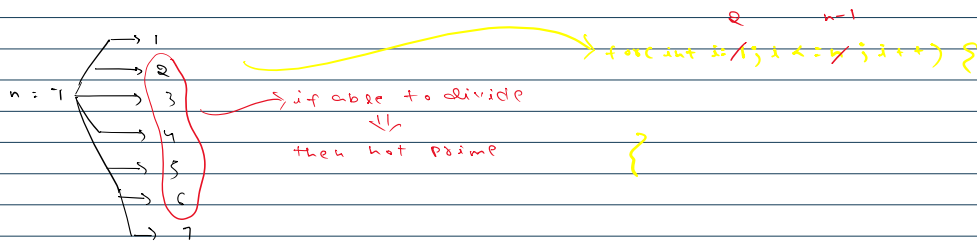
$10 \% 5 = 0$ } modulus $\Rightarrow$ rem



~~for(int i=1; i <= n; i++){~~

```
public class CheckPrime {
    public static void main(String[]
args) {
        int n = 9;
        int count = 0;
        for (int i = 2; i <= n; i++) {
            if (n % i == 0) {
                count++;
            }
        }
        if (count >= 1) {
            System.out.println("Not
```





```

}
if (count >= 1) {
    System.out.println("Not
    Prime");
} else {
    System.out.println("Prime");
}
}
}

```

A no. is prime only when it is not divisible by any no in the range of 2 to $n-1$

sum of number

$$n \rightarrow 2346 \rightarrow 2 + 3 + 4 + 6 \rightarrow 15 \rightarrow 1 + 5 = 6$$

$$2346 \% 10 = 6$$

$$\rightarrow 2346 / 10 = 234$$

$$\rightarrow 234 \% 10 = 4$$

$$\rightarrow 234 / 10 = 23$$

$$\rightarrow 23 \% 10 = 3$$

$$\rightarrow 23 / 10 = 2$$

$$\rightarrow 2$$

→ 10 - 2

→ 210 = 0